

Force

Force : Push or pull , Newton (N) $\approx$ the force it takes to hold up a small apple ($\sim 100g$) against the gravity

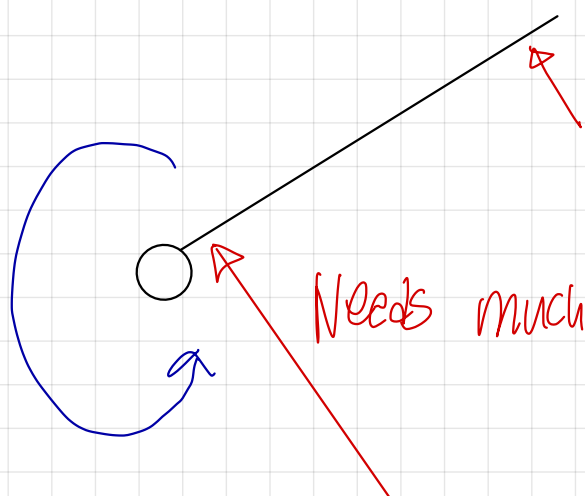
$$(kg \cdot m/s^2) \quad (kg) \quad (m/s^2)$$

$$Force = mass \times acceleration$$

$$F = m a$$

Torque

Torque : a Twisting force , the rotational version of regular push/pull force



$$(N \cdot m) \quad (N) \quad (m)$$

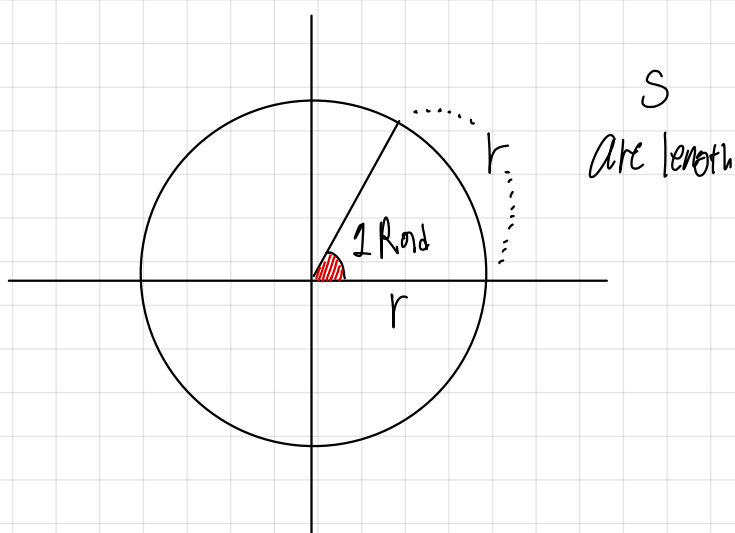
$$torque = force \times distance \text{ from pivot}$$

$$= \text{lever arm}$$

$$= \text{moment arm}$$

$$\tau = F r$$

Radian



$$(m) \quad (m) \quad (\text{unitless})$$

$$Arc \text{ length} = radius \times \text{angle in radians}$$

$$s = r \theta$$

$$\theta = \frac{s}{r}$$

↳ A full circle : $360^\circ = 2\pi$ radians

$$\theta(\text{rad}) = \theta(\text{deg}) \times \frac{2\pi}{360}$$

$$\theta(\text{deg}) = \theta(\text{rad}) \times \frac{360}{2\pi}$$

↳ π = $\frac{\text{Circumference (m)}}{\text{Diameter (m)}}$ (unitless)

of ANY Circle

$$\pi = C/d \approx 3.14$$

↳ $C = \pi \times d = \pi \times 2r = 2\pi r$

$= (2\pi) r$

If arc length is C (the whole circle) then

$$S = \theta r \Rightarrow \theta = (2\pi)$$

Angular Velocity and Acceleration

$$\text{Velocity} = \frac{\Delta \text{position (m)}}{\Delta \text{time (s)}} \quad (\text{m/s})$$

$$V = \frac{\Delta x}{\Delta t}$$

How much distance per unit time

$$\text{Angular Velocity} = \frac{\Delta \text{angle (in rad)}}{\Delta \text{time (s)}} \quad (\text{rad/s})$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

How much angle per unit time

$$\text{Acceleration} = \frac{\Delta \text{velocity (m/s)}}{\Delta \text{time (s)}} \quad (\text{m/s}^2)$$

$$a = \frac{\Delta v}{\Delta t}$$

How fast velocity changes
per unit time

$$\text{Angular acceleration} = \frac{\Delta \text{angular velocity (rad/s)}}{\Delta \text{time (s)}} \quad (\text{rad/s}^2)$$

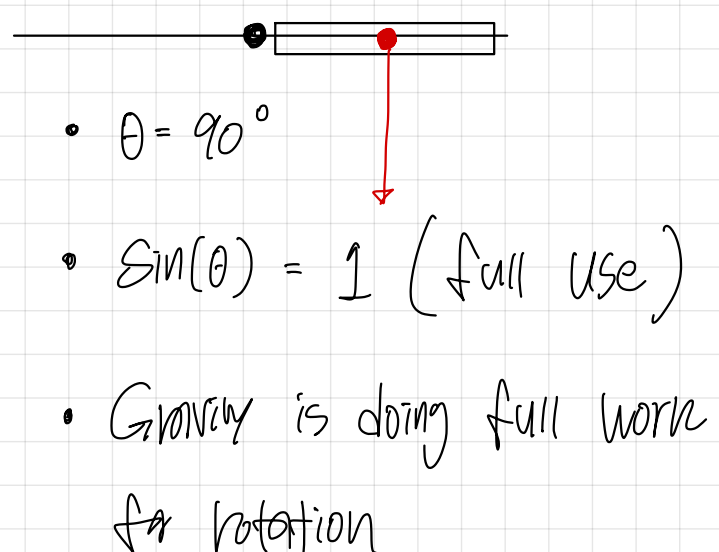
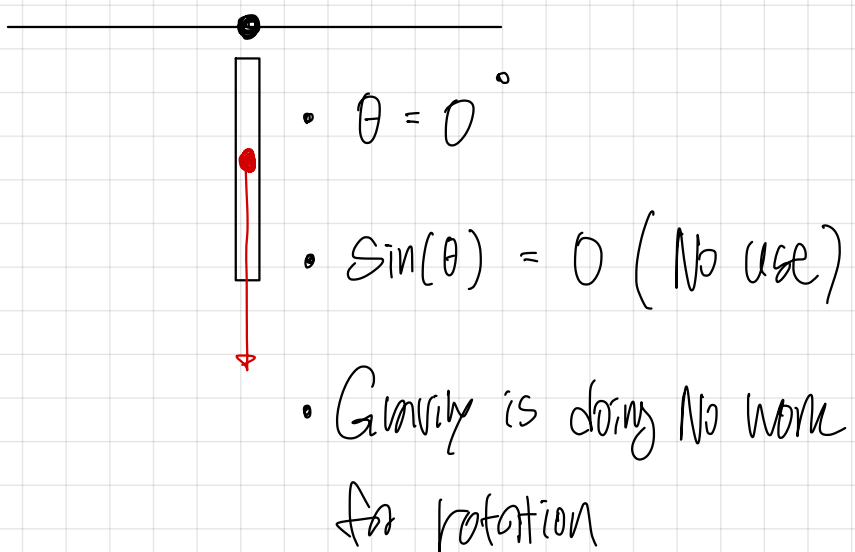
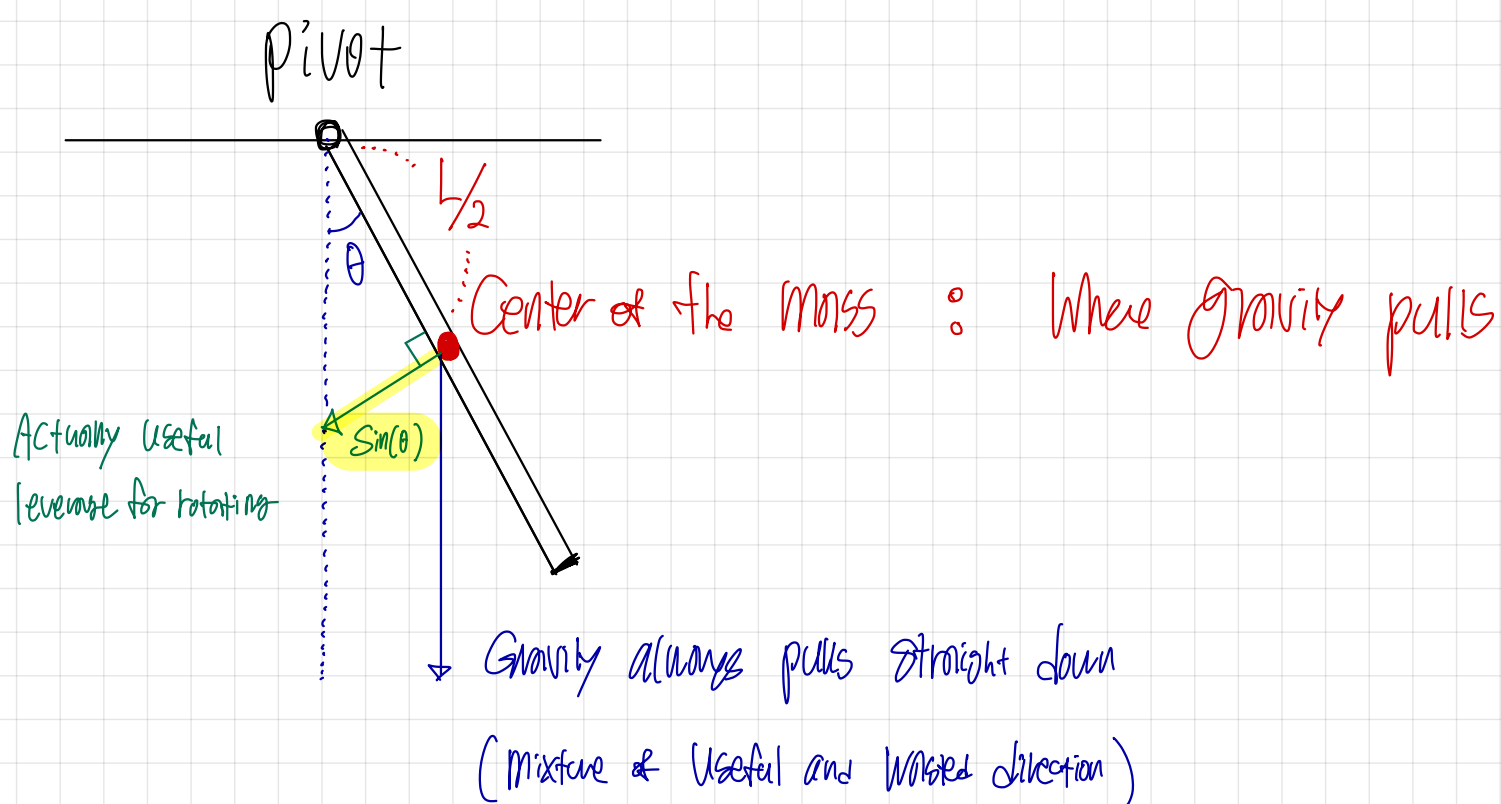
$$\alpha = \frac{\Delta \omega}{\Delta t}$$

How fast angular velocity changes
per unit time

Gravitational Torque

Gravitational Torque : the torque that gravity creates around the pivot trying to swing the pendulum

↳ torque $\tau = F \times r$ (moment arm)



Moment Arm : the perpendicular distance from the pivot to the line along with the force acts

$$\text{Moment arm} = \frac{L}{2} \times \sin(\theta)$$

(m)
(m)
(dimensionless)

$$I_{\text{gravity}} = F_{\text{gravity}} \times \text{Moment Arm}$$

(N·m)
(N)
(m)

$$= (m \times g) \times \left(\frac{L}{2} \times \sin(\theta) \right)$$

(kg)
(m/s²)
(m)
(unitless)

Damping Torque

Damping : Any force or torque that opposes the direction of current motion, grows in strength as the speed of motion increases, and is zero when there is no motion at all

1) Viscous damping (linear)

$$(N·m) \quad (N·m·s/rad) \quad (rad/s)$$

$$I = -C \times \omega$$

Smooth, proportional to speed, realistic for air resistance
or fluid-line resistance in a joint

2) Coulomb Damping (dry friction or sticktion)

(N·m) (N·m) (unitless)

$$\tau = -\mu \times \text{Sign}(\omega)$$

A constant resistance, same strength no matter how fast

3) Quadratic Damping (air resistance at high speed)

(N·m) (N·m·s²/rad²) (rad/s) (rad/s)

$$\tau = -C \times \omega \times |\omega|$$

↳ τ_{total} for the pendulum model)

$$\tau_{\text{total}} = \tau_{\text{gravity}} + \tau_{\text{viscous}} + \tau_{\text{Coulomb}}$$

Inertia

Inertia : an Object's natural resistance to having its Motion Changed (Newton's first law)

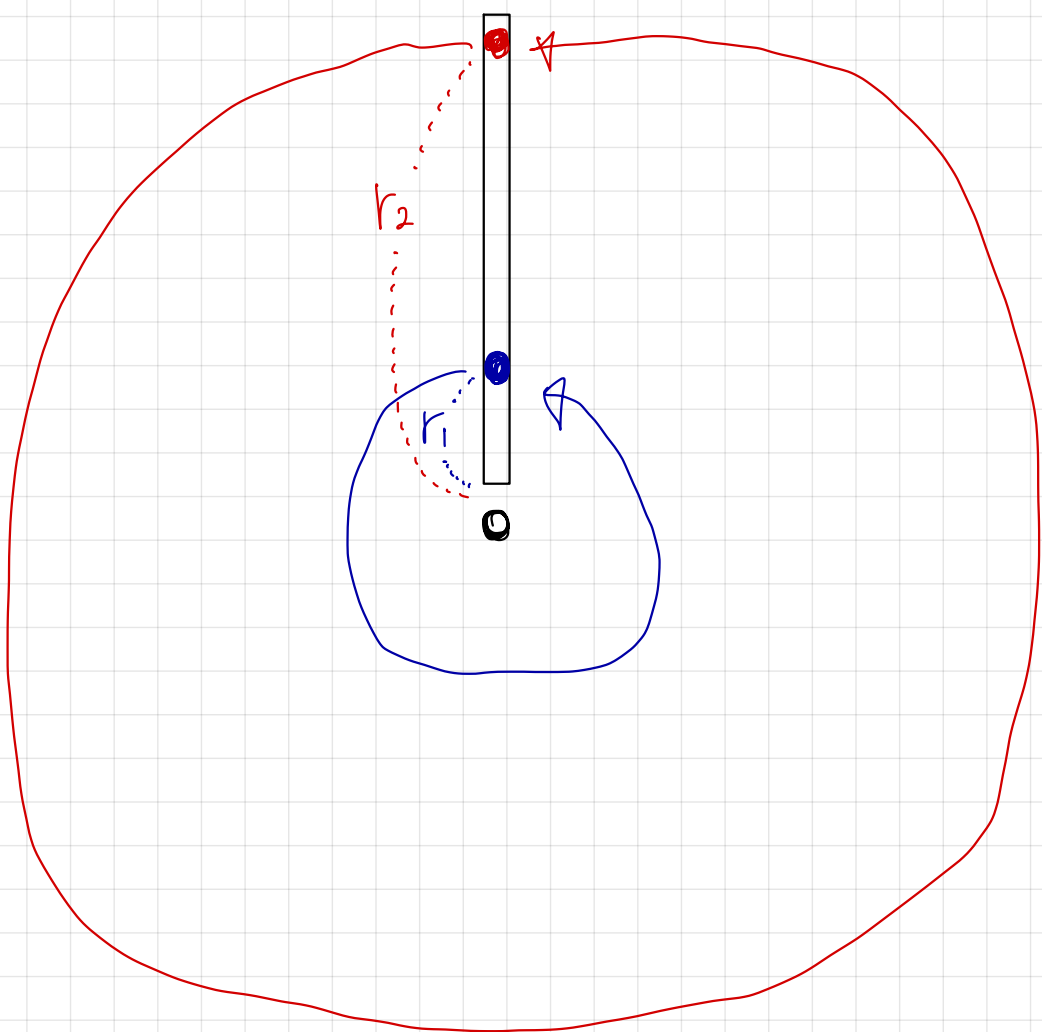
↳ Straight line motion : Inertia = Mass (kg)

$$F = \underline{m} a$$

Linear Motion (Translational Motion) Not for Rotational Motion

↳ Why Chunks ?

Why Rotational Motion Needs something More than Mass?



Both points has the same amounts of time to rotate

A point sitting far away from the pivot should travel much further thus require higher speed

⇒ more expensive to spin

$$(kg \cdot m^2) \quad (kg) \quad (m)$$

$$I = m \cdot r^2$$

0) If we push a mass on the end of a rod to make it spin faster, how much push is needed in terms of angular quantities?

$$1) \quad s = r \times \theta \quad \Rightarrow \quad \frac{ds}{dt} = r \times \underbrace{\frac{d\theta}{dt}}_{\omega : \text{angular velocity}}$$

↳ the distance if the mass sweeps through some angle θ

2) Velocity is defined as how fast distance changes over time

$$v = \frac{ds}{dt} = r \omega, \quad \frac{dv}{dt} = r \times \underbrace{\frac{d\omega}{dt}}_{\alpha : \text{angular acceleration}}$$

3) Acceleration : how fast velocity changes over time

$$a = \frac{dv}{dt} = r \alpha$$

4) Newton's second Law

$$F = m \times a = m \times r \times \alpha$$

5) Torque

$$L = F \times r = m \times r \times \alpha \times r = \underbrace{m r^2}_{I} \alpha$$

6) General Rule for Rotation (N2L for rotation)

$$\therefore L_{\text{net}} = I \times \alpha$$

$$\hookrightarrow L_1 = m_1 \times r_1^2 \times \alpha$$

$$L_2 = m_2 \times r_2^2 \times \alpha$$

$\vdots$

$$L_n = m_n \times r_n^2 \times \alpha$$

$$L_{\text{total}} = L_1 + L_2 + \dots + L_n = \alpha \times \underbrace{\sum m_i r_i^2}_I$$

(kg · m²)

(kg · m²)

$$\therefore I = \frac{L}{\alpha} = m r^2$$

↳ Discrete Chunks) $I = \sum m_j r_j^2$

Continuous Chunk) $I = \int r^2 dm = \frac{1}{3} m L^2$

i) mass per unit length

Total mass of the rod : m [kg]

Total length of the rod : L [m]

mass per unit length : m/L [kg/m]

ii) tiny slice

dr : a very small length

⇒ How much mass in that slice?

$$dm = \text{mass per unit length} \times dr$$

$$dm = \left(\frac{m}{L} \right) \times dr$$

↳ One chunk mass

iii) Contribution

Contribution depends on the moment arm r

$$\text{Contribution} = dm \times r^2$$

$$= \left(\frac{m}{L} \right) dr \times r^2$$

↳ One slice's contribution

iv) Total Contribution

$$I = \int_0^L r^2 \left(\frac{m}{L} \right) dr$$

$$= \frac{m}{L} \int_0^L r^2 dr$$

$$= \frac{m}{L} \left(\frac{r^3}{3} \Big|_{r=L} - \frac{r^3}{3} \Big|_{r=0} \right)$$

$$= \frac{m}{L} \cdot \frac{L^3}{3}$$

$$I = \frac{1}{3} m L^2$$

• Interpretation) $J = I / \alpha$

J is the amount of torque required to produce one unit of angular acceleration

$\Rightarrow$ If $J = 0.083 \text{ kg} \cdot \text{m}^2$ for the pendulum, to make this pendulum accelerate angularly at 1 rad/s^2 , we need exactly $0.083 \text{ N} \cdot \text{m}$ of torque